

6 (a) ohm is volt per ampere or volt/ampere B1 [1]

(b) (i)  $R = \rho l / A$  B1

$$R_P = 4\rho(2l)/\pi d^2 \text{ or } 8\rho l/\pi d^2 \text{ or } R_Q = \rho l/\pi d^2$$

or

ratio idea e.g. length is halved hence  $R$  halved and diameter is halved hence  $R$  is  $1/4$  C1

$$\begin{aligned} R_Q (= 4\rho l/\pi 4d^2) &= \rho l/\pi d^2 \\ &= R_P/8 \\ &= (12/8) = 1.5 \, \Omega \end{aligned}$$

A1 [3]

(ii) power =  $I^2 R$  or  $V^2/R$  or  $VI$  C1

$$= (1.25)^2 \times 12 + (10)^2 \times 1.5 \text{ or } (15)^2/12 + (15)^2/1.5 \text{ or } 15 \times 11.25$$
 C1

$$= (18.75 + 150 =) 170 \text{ (168.75) W}$$
 A1 [3]

(iii)  $I_P = (15/12 =) 1.25 \text{ (A)}$  and  $I_Q = (15/1.5 =) 10 \text{ (A)}$  C1

$$V_P/V_Q = I_P n A_Q e / I_Q n A_P e \text{ or } (1.25 \times \pi d^2)/(10 \times \pi d^2/4)$$
 C1

$$= 0.5$$
 A1 [3]